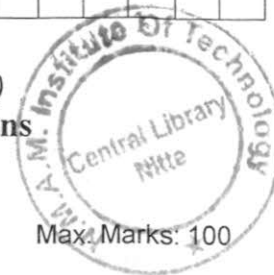


NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May - June 2023

PH1001-1 – ENGINEERING PHYSICS



Duration: 3 Hours

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

List of constants: Velocity of light, $c=3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h=6.63 \times 10^{-34} \text{ Js}$, Electron mass, $m=9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e=1.602 \times 10^{-19} \text{ C}$, Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, Boltzmann constant, $k=1.38 \times 10^{-23} \text{ J/K}$, Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- A wave packet is used to represent**
 A) A light wave
 B) a stationary wave
 C) Matter wave
 D) a transverse wave
- The wave function is an acceptable wave function if it is**
 A) finite everywhere
 B) continuous everywhere
 C) single valued everywhere
 D) having all these properties
- The normalized wave function must have _____ norm**
 A) infinite
 B) zero
 C) finite
 D) complex
- The energy corresponding to the lowest permitted energy level for a particle in an infinite potential well is called**
 A) Excited energy
 B) Zero-point energy
 C) Metastable state energy
 D) None of these
- The smallest portion of a crystal which when repeated in different directions generates the entire crystal is called:**
 A) Lattice points
 B) Crystal lattice
 C) Unit cell
 D) None of the mentioned
- In a triclinic crystal:**
 A) $a = b = c, \alpha = \beta = \gamma \neq 90^\circ$
 B) $a \neq b = c, \alpha = \beta = \gamma = 90^\circ$
 C) $a \neq b \neq c, \alpha \neq \beta \neq \gamma \neq 90^\circ$
 D) $a \neq b \neq c, \alpha = \beta = 90^\circ \neq \gamma \neq 90^\circ$
- If R is the radius of the atom in a crystal, crystallizing in the simple cubic structure, then the nearest neighbor distance is**
 A) $R/2$
 B) $4R$
 C) $2R$
 D) $2\sqrt{2}R$
- X-rays have**
 A) Short wavelength
 B) high frequency
 C) both Short wavelength and high frequency
 D) low frequency
- Hall Effect is clearly visible in _____**
 A) Pure conductors
 B) Semiconductors
 C) Super conductors
 D) Metals
- The number of electrons in a semiconductor is 10^{20} . Then the Hall coefficient is**
 A) 0.625
 B) 0.0625
 C) 6.25
 D) 62.5
- When a pentavalent impurity is added to a pure semiconductor, it becomes**
 A) An insulator
 B) An intrinsic semiconductor
 C) p-type semiconductor
 D) n-type semiconductor

12. A trivalent impurity has _____ Valence electrons
 A) 3 B) 5
 C) 4 D) 6
13. The magnetic susceptibility (χ) in superconductor is
 A) positive B) zero
 C) negative D) infinity
14. In superconductors the temperature below which conductivity of a material becomes infinite is called
 A) Critical temperature B) Absolute temperature
 C) Mean temperature D) Crystallisation temperature
15. Photodiode is operated in reverse Bias because
 A) Photodiode does not work in forward bias B) The fractional change in the reverse current when photons strike the diode is substantially bigger than the fractional change in the forward current.
 C) Current response is linear in reverse bias D) Photodiode will get damaged if used in forward bias
16. Intense magnetic field in a coil can be produced using a
 A) Conductor with large number of turns and large current B) Type-I Superconductor with large number of turns and large current
 C) Type-II Superconductor with large number of turns and large current D) Type-II Superconductor with large number of turns and current below critical current
17. The need to achieve population inversion is
 A) To excite most of the atoms B) To bring most of the atoms to ground state
 C) To achieve stable condition D) To reduce the time of production of laser
18. In a laser, the mirrors are there on either side of the device
 A) In order to filter the heat & leave out only the light. B) To protect the eyes of the observer.
 C) To save energy of the excitation source D) So that same photons continue to cause further batches of stimulated transitions
19. Having cladding around the core is preferred to coating the core with a reflecting material(silvering) because
 A) Silvering is not economical B) Coating may get affected by abrasions
 C) Coated material may undergo chemical changes in the field conditions and thus become dull D) The total internal reflection at the core-cladding interface is superior to the one by any coated material
20. Attenuation means
 A) Amplification of signal strength B) Division of signal strength
 C) Loss of signal strength D) Tuning of signal

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are matter waves? Write any three of their characteristics.	4	L2	1	1, 2
	b) Obtain an expression for one dimensional time independent Schrodinger's wave equation.	8	L2	1	1, 2
	c) An electron is confined to move between two rigid walls separated by 20 Å. Calculate the ground state energy of an electron and its de-Broglie wavelength (assume the potential to be zero).	4	L3	1	1, 2
2.	a) Explain the following terms: (i) Probability density (ii) Normalization of a wave function.	4	L2	1	1, 2
	b) What is atomic packing factor? Determine the atomic packing factor of base centered cubic (BCC) lattice by calculating number of atoms per unit cell and obtaining the relation between atomic radius and lattice constant.	8	L2	2	1, 2
	c) Copper has FCC structure of atomic radius 0.1278 nm. Calculate the interplanar spacing for (3 2 1) plane.	4	L3	2	1, 2

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|-------|---|---|----|---|------|
| 3. a) | What are Miller indices? With an example explain the procedure to find Miller indices. | 4 | L2 | 2 | 1, 2 |
| b) | Explain the Origin of continuous X-rays. | 8 | L2 | 2 | 1, 2 |
| c) | A monochromatic X ray beam of wavelength 0.7 Å undergoes first-order Bragg reflection from the plane (3 0 2) of a cubic crystal at a glancing angle of 35°. Calculate the lattice constant. | 4 | L3 | 2 | 1, 2 |

Unit – II

- | | | | | | |
|-------|---|---|----|---|------|
| 4. a) | Write any four differences between intrinsic and extrinsic semiconductors. | 4 | L1 | 3 | 1, 2 |
| b) | Derive an expression for the electrical conductivity of an intrinsic semiconductor in terms of carrier concentration and carrier mobilities. Comment on the effect of temperature on conductivity. | 8 | L2 | 3 | 1, 2 |
| c) | Mobilities of electrons and holes in a sample of intrinsic germanium at 300 K are $0.36 \text{ m}^2 \text{V}^{-1} \text{S}^{-1}$ and $0.17 \text{ m}^2 \text{V}^{-1} \text{S}^{-1}$ respectively. If the resistivity of the specimen is $2.12 \Omega \text{m}$, compute the intrinsic carrier density. | 4 | L3 | 3 | 1, 2 |
| 5. a) | Explain the effect of temperature on the Fermi level in an n-type semiconductor. | 4 | L2 | 3 | 1, 2 |
| b) | What is Hall effect? Explain the production of Hall field and obtain an expression for the carrier concentration of an n - type semiconductor and Hall coefficient. | 8 | L2 | 3 | 1, 2 |
| c) | The Hall coefficient of a specimen of a doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3 \text{C}^{-1}$ and its resistivity is found to be $8.93 \times 10^{-3} \Omega \text{m}$. Calculate charge carrier concentration and electron mobility at room temperature. | 4 | L3 | 3 | 1, 2 |
| 6. a) | Write a note on BCS theory for superconductors. | 4 | L2 | 4 | 1, 2 |
| b) | What is superconductivity? Explain the magnetic behavior of Type - I and Type – II superconductors. | 8 | L3 | 4 | 1, 2 |
| c) | Superconducting tin has a critical magnetic field of 217 gauss at 2 K. If the critical temperature for superconducting transition for tin is 3.7 K, find the critical magnetic field at 3 K. | 4 | L3 | 4 | 1, 2 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|------|
| 7. a) | What is a laser? Explain the principle of stimulated emission of radiation. | 4 | L2 | 5 | 1, 2 |
| b) | With neat diagrams, describe the construction and working of He-Ne laser. | 8 | L2 | 5 | 1, 2 |
| c) | A He-Ne laser has an output power of 10mw and emits light at a wavelength of 632.8 nm. How many photons are emitted by the laser in each minute? | 4 | L3 | 5 | 1, 2 |
| 8. a) | What is an optical fiber? Explain the principle on the basis of which optical transmission is achieved through a fiber. | 4 | L2 | 5 | 1, 2 |
| b) | What are the different modes of propagation possible in optical fibers? Explain in detail different multi mode fibers with necessary diagrams. | 8 | L2 | 5 | 1, 2 |
| c) | The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33. | 4 | L3 | 5 | 1, 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
